SCE-013 - FROM PRACTITIONER INTERVIEWS TO STRUCTURAL DISCOVERY
Why Collective Learning May Matter More Than Literature Review in the AI Era

Sizhe Tan
with chatGPT (OpenAI)
2026-06-19

Repository: https://github.com/sizhet/Structural-Cognitive-Emergence-SCE

Abstract
For centuries, scientific progress has relied heavily on literature review.
Researchers study existing papers, identify unresolved questions, and gradually extend the frontier of knowledge.
However, the rapid development of Artificial Intelligence is exposing a new challenge: many of the most important questions emerge faster than the publication cycle itself.
In highly dynamic domains, critical insights often appear first in practitioner experiences, engineering bottlenecks, deployment failures, and research interviews rather than in formal publications.
This article proposes a complementary methodology:
Practitioner Interview → Gap Detection → Why Generation → Collective Learning → Structural Discovery
We argue that in rapidly evolving fields, practitioner interviews can serve as a rich source of discovery signals because they expose unresolved structural gaps rather than already-organized knowledge. Combined with collective reasoning and structural analysis, these gaps can become the starting point for new theories, frameworks, and scientific breakthroughs.

1. Introduction
Traditional scientific research is largely organized around published knowledge.
A common workflow looks like:
Literature Review
↓

Known Problems
↓
Incremental Improvement
↓
New Publication
This model has been extraordinarily successful.
However, it assumes that the most valuable information already exists in published form.
In rapidly evolving fields such as Artificial Intelligence, this assumption is becoming increasingly questionable.
Many critical questions emerge long before formal explanations appear.
Often they first surface as:

- engineering frustrations,
- unexplained behaviors,
- deployment failures,
- surprising observations,
- practitioner interviews,
- open discussions among researchers.

These signals frequently contain the seeds of future discoveries.

2. Papers Provide Answers

Most papers are designed to communicate solutions.
A published paper typically explains:

- what was attempted,
- what worked,
- what failed,
- what conclusions were reached.

In other words:
Paper
=
Organized Knowledge
Papers are excellent tools for preserving and transmitting understanding.
They help the community consolidate discoveries.
However, by the time knowledge becomes a paper, much of the original uncertainty has already been filtered out.
The gaps have often been hidden behind the final narrative.

3. Interviews Expose Gaps

Practitioner interviews operate differently.
Researchers and engineers often say things such as:

We do not fully understand why this happens.
The model suddenly acquires this capability.
This behavior surprised us.
We still cannot explain this phenomenon.
These statements are not answers.
They are gaps.
Unlike papers, interviews frequently expose:

- unresolved contradictions,
- unexplained observations,
- recurring bottlenecks,
- emerging puzzles.

Therefore:
Interview
=
Gap Exposure
And gaps are precisely where discovery begins.

4. Gap Generates Why

Structural Cognitive Emergence (SCE) suggests a simple but powerful chain:
Gap
↓
Why
↓
Exploration
↓
New Concept
A gap creates tension.
The tension generates questions.
Questions generate curiosity.
Curiosity initiates exploration.
Exploration produces new concepts and theories.
From this perspective, the value of interviews is not that they provide answers.
Their value is that they reveal the locations of gaps.

5. The Discovery Pipeline

Many important discoveries may follow a process closer to:
Practitioner Interview
↓
Gap Detection

↓
Why Generation
↓
Collective Discussion
↓
Structural Abstraction
↓
Theory Formation
This process differs fundamentally from traditional literature-driven research.
The objective is not merely to absorb existing knowledge.
The objective is to identify emerging structural gaps before they become obvious to everyone else.

6. Collective Learning as a Discovery Engine

The next crucial step is discussion.
A gap alone does not guarantee discovery.
Different participants bring different perspectives:

- engineers,
- scientists,
- domain experts,
- system designers,
- AI systems.

Each possesses a different Trigger Network.
When multiple Trigger Networks interact, new connections emerge.
The same observation can activate entirely different chains of reasoning.
This process can be represented as:
Gap
↓
Multiple Perspectives
↓
Trigger Activation
↓
New Connections
↓
Why Expansion
↓
Structural Insight
Collective Learning is therefore not simply shared knowledge.
It is shared gap exploration.

7. From Knowledge Consumption to Gap Mining

Traditional research often emphasizes knowledge consumption:

Read
↓
Understand
↓
Extend

An alternative methodology emphasizes gap mining:

Observe
↓
Detect Gap
↓
Ask Why
↓
Build Structure

In rapidly changing fields, this approach may become increasingly important.

The most valuable information may not be the answer already written in a paper.

It may be the unanswered question hidden inside a conversation.

8. AI and the Rise of Collective Discovery

Artificial Intelligence may accelerate this transformation.

Large language models can:

- summarize interviews,
- compare perspectives,
- identify recurring themes,
- highlight contradictions,
- amplify gap detection,
- support collective reasoning.

As a result, scientific discovery may gradually evolve from:

Individual Research

toward

Human–AI Collective Discovery

where humans provide experience, intuition, and context while AI helps identify patterns, connections, and emerging gaps.

9. A New Research Loop

The emerging discovery cycle may look like:

Interview
↓

Trigger
↓
Gap
↓
Why
↓
Collective Learning
↓
Structural Discovery
↓
Theory
↓
New Gap
This loop closely resembles the process of cognitive emergence itself.
Knowledge no longer appears as isolated answers.
It emerges from repeated interaction with unresolved gaps.

10. Conclusion

Literature review remains indispensable.
Science cannot progress without preserving and understanding existing knowledge.
However, in the AI era, literature review may no longer be sufficient by itself.
Practitioner interviews expose something equally important:
the frontier of uncertainty.
They reveal where current understanding breaks down.
They expose gaps before theories exist.
When combined with collective learning and structural reasoning, these gaps can become the source of entirely new research directions.
From this perspective, the future of discovery may depend not only on reading more papers.
It may depend on learning how to find better gaps.

Key Insight

Papers preserve knowledge.
Interviews expose gaps.
Gaps generate Why.
Why drives discovery.
Collective Learning transforms gaps into new theories.